Nandana Narayan Das

+91 7907382133 | nandanadas007@gmail.com | github.com/nandana-das | linkedin.com/in/nandana-narayan-das

OBJECTIVE

M.Tech Artificial Intelligence and Data Science student with experience in machine learning, predictive modelling, computer vision, retrieval-augmented generation, customer analytics, and data visualisation. Skilled in analysing real-world datasets, developing and evaluating AI/ML pipelines, and communicating findings through clear reports and dashboards. Interested in applying data science and AI to practical problems across research and industry.

TECHNICAL SKILLS

Data Science & Analytics: Python, Pandas, NumPy, SQL, EDA, predictive modelling, customer segmentation, statistical analysis
ML / DL: scikit-learn, TensorFlow, PyTorch, Keras, LSTM, CNN, YOLO, model evaluation and benchmarking
AI / GenAI: LangChain, LangGraph, RAG pipelines, FAISS, prompt engineering, Groq API, Gemini API
Visualisation & BI: Matplotlib, Power BI, Streamlit, data storytelling, dashboard reporting
Tools: Git, GitHub, Jupyter Notebook, Google Colab, spaCy, NLTK

PROJECTS

Rockfall-Related Feature Detection in Chandrayaan-2 OHRC Imagery | *Python · PyTorch · YOLO · Computer Vision · Object Detection*

- Developed a computer-vision pipeline for boulder-like feature detection in Chandrayaan-2 OHRC imagery, including image preprocessing, annotation conversion, model training, and inference. The best reported source-domain result was F1-score 0.6275 and mAP@0.5 0.6463 on a 262-image held-out test set.
- Compared YOLO model variants using confidence analysis and held-out evaluation; prepared workflows for future target-domain evaluation. Target OHRC tiles remain unlabeled, so target-domain accuracy has not been established.

KG-RAG-ISRO — Knowledge-Graph-Augmented Retrieval-Augmented Generation | *Python · FAISS · spaCy · Knowledge Graphs · Local LLMs*

- Developed an ISRO-domain question-answering system combining FAISS dense retrieval with relation-aware knowledge-graph retrieval; implemented model-token-aware prompt budgeting and regression tests to preserve high-priority evidence within context windows.
- Evaluated against vanilla dense RAG on a frozen benchmark — reported higher mean correctness and fewer unsupported claims, demonstrating structured experimentation and rigorous evaluation methodology.

E-Commerce Customer Behaviour Analysis — REES46 Cosmetics Shop | *Python · Pandas · Power BI · EDA · Business Intelligence*

- Analysed approximately 3.5M clickstream events to examine customer journeys across product views, cart activity, and purchases — identifying key drop-off points and conversion patterns directly relevant to product intelligence.
- Developed customer behaviour, segmentation, and temporal analyses, translating findings into business KPIs and actionable recommendations for retention and revenue optimisation.
- Built a Python-to-Power BI reporting workflow with processed datasets and an interactive dashboard for stakeholder-ready decision support.

Smart City Traffic AI — Bengaluru | *Python · LSTM · TomTom API · Fuzzy Cognitive Maps · FastAPI*

- Developed a traffic forecasting prototype using TomTom data and location-specific LSTM models for congestion prediction; integrated a Fuzzy Cognitive Map layer for interpretable decision-making and documented an evaluation workflow comparing LSTM forecasts against a naive baseline.

PUBLICATIONS

Explainable AI for Disease Detection and Body Constitution Analysis Using Tongue Imaging — IEEE Xplor, ICTEST 2025 View on IEEE
Physics-Informed Explainable Multimodal Deep Learning for Satellite-Based Air Pollution Source Attribution in Delhi NCR — *IEEE Xplor, ICTEST 2026* , View on IEEE

CERTIFICATIONS

Oracle Certified Java Foundations Associate · IBM AI Foundation · Microsoft Generative AI Essentials · KNIME Data Science Certificate

EDUCATION

M.Tech in Artificial Intelligence and Data Science *2025 – 2027*
Alliance School of Advanced Computing, Alliance University — Bangalore
B.Tech in Computer Science and Engineering (AI)*2021 – 2025*
Adi Shankara Institute of Engineering and Technology — Kalady, Kerala